# Homework 6

```
A = readmatrix("HW6_A.xlsx");
B = [2; 0; 1.5; 0];
C = [1 0 2 0];
D = 0;

AA = [A B; C 0] ;
bb = [0; 0; 0; 0; 1] ;
n = size(A,1);   % state dimension = 4
```

## part (a) fall state regulator and estimator

```
% Open-loop poles
ol_poles = eig(A);
fprintf('Open-loop poles:\n'); disp(ol_poles)
```

```
Open-loop poles:
  -1.0000 + 9.0000i
  -1.0000 - 9.0000i
  -4.0000 + 0.0000i
 -12.0000 + 0.0000i
```

The open-loop eignvalues are $\lambda(A) = \{-1 \pm 9i, -4, -12\}\ \mathrm{rad}/s$

Since the state of the plant can not be measuered, control law we use estimated state $\hat{x}$ from the estimator.

$u = -K\hat{x}$

To achieve undamping, we choose a damping ratio of $\zeta = 0.7$based on Figure 3.19 (Frankin). We want the settling time less than 1 seconds, we pick the situation of 5%

Settling time $e^{-\zeta\omega_n t_s} = 0.05 \rightarrow t_s = \frac{3}{\zeta\ \omega_n} = \frac{3}{\sigma}$

so $\zeta = \sigma \geq 3.333$. For measure of safety, we chose $\zeta\omega_n = \sigma = 5$. since $\zeta = 0.7, \omega_n = 7.07$

$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -5 \pm 5j$

the dominant poles we choose is $\{-5 \pm 5j\}$. Two non-dominant poles are placed at $\{-10, -15\}$.
$(2-4 \times \text{faster than the dominant poles})$.

```
% Regulator poles (well into LHP, good damping)
reg_poles = [-5+5i, -5-5i, -10, -15];
K = place(A, B, reg_poles);
fprintf('State feedback gain K:\n'); disp(K);
```

```
State feedback gain K:
    8.8579    0.7343   -0.4771    6.3417
```

```
fprintf('Closed-loop (regulator) poles:\n'); disp(eig(A - B*K));
```

```
Closed-loop (regulator) poles:
 -15.0000 + 0.0000i
  -5.0000 + 5.0000i
  -5.0000 - 5.0000i
 -10.0000 + 0.0000i
```

```
% Observer poles (3-7x faster than regulator)
obs_poles = [-20+8i; -20-8i; -25+5i; -25-5i];
L = place(A', C', obs_poles)';
fprintf('Observer gain L:\n'); disp(L);
```

```
Observer gain L:
   14.2225
 -165.7479
   28.8887
  165.6131
```

```
fprintf('Observer (A-LC) poles:\n'); disp(eig(A - L*C));
```

```
Observer (A-LC) poles:
 -25.0000 + 5.0000i
 -25.0000 - 5.0000i
 -20.0000 + 8.0000i
 -20.0000 - 8.0000i
```

## part (b) reference tracking system

The estimator : $\dot{\hat{x}} = (A - \mathrm{LC} - \mathrm{BK})\hat{x} + \mathrm{Ly} + B\bar{N}r$

control law : $u = -K\hat{x} + \bar{N}r$.

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \underbrace{\begin{bmatrix} A & -\mathrm{BK} \\ C & A - \mathrm{BK} - \mathrm{LC} \end{bmatrix}}_{A_{\mathrm{cl},b}} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \underbrace{\begin{bmatrix} B\bar{N} \\ B\bar{N} \end{bmatrix}}_{B_{\mathrm{cl},b}} r$$

$$y = \underbrace{\begin{bmatrix} C & 0 \end{bmatrix}}_{C_{\mathrm{cl},b}} \begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix}$$

```
xx = AA \ bb;

Nx = xx(1:4);
Nu = xx(5);
N_bar = Nu+ K * Nx;
display(Nx);
```

```
Nx = 4×1
   -0.1947
   -0.1472
    0.5973
    0.4654
```

```
display(Nu);
```

```
Nu =
0.9208
```

```
display(Nbar);
```

```
Nbar =
0.8983
```

```
Acl_b = [A,        -B*K;
         L*C,    A-L*C-B*K];

Bcl_b = [B*N_bar;
         B*N_bar];

Ccl_b = [C, zeros(1,n)];
Dcl_b = 0;

sys_b = ss(Acl_b, Bcl_b, Ccl_b, Dcl_b);  %delete the simicolon to show the
system
fprintf('Part (b) closed-loop poles:\n'); disp(eig(Acl_b));
```

```
Part (b) closed-loop poles:
  -5.0000 + 5.0000i
  -5.0000 - 5.0000i
 -25.0000 + 5.0000i
 -25.0000 - 5.0000i
 -10.0000 + 0.0000i
 -15.0000 + 0.0000i
 -20.0000 + 8.0000i
 -20.0000 - 8.0000i
```

## part (c) robust tracking

$\dot{x_I} = \mathrm{Cx} - r(= e)$

$\dot{x} = \mathrm{Ax} + \mathrm{Bu} = \mathrm{Ax} + B(-K_1 x_I - K_0 \widehat{x})$

$u = -K_1 x_I - K_0 \widehat{x}$

$\dot{\widehat{x}} = A\widehat{x} + \mathrm{Bu} + L(y - C\widehat{x}) = (A - \mathrm{BK}_0 - \mathrm{CL})\widehat{x} - \mathrm{BK}_1 x_I + \mathrm{CLx}$

$$\begin{bmatrix} \dot{x_I} \\ \dot{x} \\ \dot{\widehat{x}} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & C & 0 \\ -\mathrm{BK}_1 & A & -\mathrm{BK}_0 \\ -\mathrm{BK}_1 & \mathrm{LC} & A - \mathrm{BK}_0 - \mathrm{LC} \end{bmatrix}}_{A_{\mathrm{aug}}} \begin{bmatrix} x_I \\ x \\ \widehat{x} \end{bmatrix} + \underbrace{\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}}_{B_{\mathrm{aug}}} r$$

$$y = \underbrace{[0 \quad C \quad 0]}_{C_{\mathrm{aug}}} \begin{bmatrix} x_I \\ x \\ \widehat{x} \end{bmatrix}$$

```
A_ext = [0,  C ;
         zeros(n,1), A];
B_ext = [0; B];
C_ext = [0, C];
fprintf('Extended system controllability rank: %d (should be %d)\n',
rank(ctrb(A_ext, B_ext)), n+1);
```

```
Extended system controllability rank: 5 (should be 5)
```

```
% Regulator poles for extended system (5 poles needed)
reg_poles_ext = [-5+5i, -5-5i, -10, -15, -20];
K_ext = place(A_ext, B_ext, reg_poles_ext);

K1 = K_ext(1);          % integrator gain (scale)
K0 = K_ext(2:end);      % state-feedback gain (1×4)

fprintf('K1 (integrator gain) = %.4f\n', K1);
```

```
K1 (integrator gain) = 35.0908
```

```
fprintf('K0 (state gains) = [%s]\n', num2str(K0));
```

```
K0 (state gains) = [12.05      6.93785      8.60006      4.52909]
```

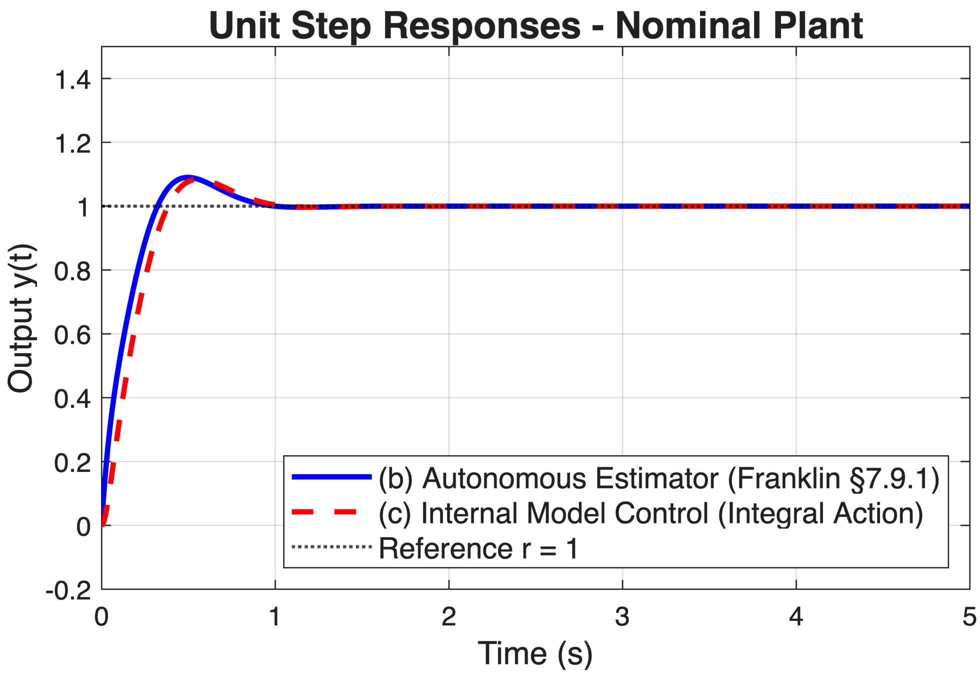
```
% Observer for original plant (same as part a)
% u = -K_0*x_hat - K_1*x_i, with x_hat from observer

% Full augmented state for simulation: z = [x_I; x; x_hat]  (9 states)
A_aug_c = [ 0,      C,              zeros(1,n)  ;
           -B*K1,   A,              -B*K0       ;
           -B*K1,   L*C,    (A - L*C - B*K0)   ];

B_aug_c = [-1; zeros(n,1); zeros(n,1)];
C_aug_c = [0, C, zeros(1,n)];

sys_c = ss(A_aug_c, B_aug_c, C_aug_c, 0)
```

```
sys_c =

  A =
             x1         x2         x3         x4         x5         x6         x7         x8         x9
   x1         0          1          0          2          0          0          0          0          0
   x2    -70.18     -5.778     -3.528     -5.788   -0.06132      -24.1     -13.88      -17.2     -9.058
   x3         0      12.16     -6.094   -0.04245      3.212          0          0          0          0
   x4    -52.64      11.16       7.66     -5.953      11.76     -18.07     -10.41      -12.9     -6.794
   x5         0      8.939      3.189      3.835    -0.1745          0          0          0          0
   x6    -70.18      14.22          0      28.45          0      -44.1      -17.4     -51.43      -9.12
   x7         0     -165.7          0     -331.5          0      177.9     -6.094      331.5      3.212
   x8    -52.64      28.89          0      57.78          0      -35.8     -2.746     -76.63      4.971
   x9         0      165.6          0      331.2          0     -156.7      3.189     -327.4    -0.1745

  B =
        u1
```

```
    x1  -1
    x2   0
    x3   0
    x4   0
    x5   0
    x6   0
    x7   0
    x8   0
    x9   0

  C =
        x1  x2  x3  x4  x5  x6  x7  x8  x9
    y1   0   1   0   2   0   0   0   0   0

  D =
        u1
    y1   0

Continuous-time state-space model.
Model Properties
```

## part (d) Unit step response

```
t = linspace(0, 5, 1000);

[y_b, t_b] = step(sys_b, t);
[y_c, t_c] = step(sys_c, t);

figure('Name','Part (d): Step Responses - Nominal
Plant','NumberTitle','off');
plot(t_b, y_b, 'b-',  'LineWidth', 2); hold on;
plot(t_c, y_c, 'r--', 'LineWidth', 2);
yline(1, 'k:', 'LineWidth', 1.2);
grid on;
xlabel('Time (s)', 'FontSize', 12);
ylabel('Output y(t)', 'FontSize', 12);
title('Unit Step Responses - Nominal Plant', 'FontSize', 14);
legend('(b) Autonomous Estimator (Franklin §7.9.1)', ...
       '(c) Internal Model Control (Integral Action)', ...
       'Reference r = 1', ...
       'Location', 'southeast', 'FontSize', 11);
xlim([0 5]); ylim([-0.2 1.5]);
```




```
% Steady-state values
fprintf('Steady-state y_b(5s) = %.4f\n', y_b(end));
```

```
Steady-state y_b(5s) = 1.0000
```

```
fprintf('Steady-state y_c(5s) = %.4f\n', y_c(end));
```

```
Steady-state y_c(5s) = 1.0000
```

## part (e) Robustness: Aged Plant

```
aging_factor = 1.50;          % 50 % increase in A
A_aged = aging_factor * A;

% --- (b) with aged plant: estimator/regulator/Nbar all use original A ---
Acl_b_aged = [A_aged,    -B*K;
              L*C,       A - L*C - B*K];
Bcl_b_aged = [B*Nbar;
              B*Nbar];
Ccl_b_aged = [C, zeros(1,n)];
sys_b_aged = ss(Acl_b_aged, Bcl_b_aged, Ccl_b_aged, 0);

fprintf('\n=== (e) Aged-plant closed-loop poles ===\n');
```

```
=== (e) Aged-plant closed-loop poles ===
```

```
fprintf('Part (b) aged poles:\n'); disp(eig(Acl_b_aged));
```

```
Part (b) aged poles:
 -45.5923 +18.5940i
 -45.5923 -18.5940i
```

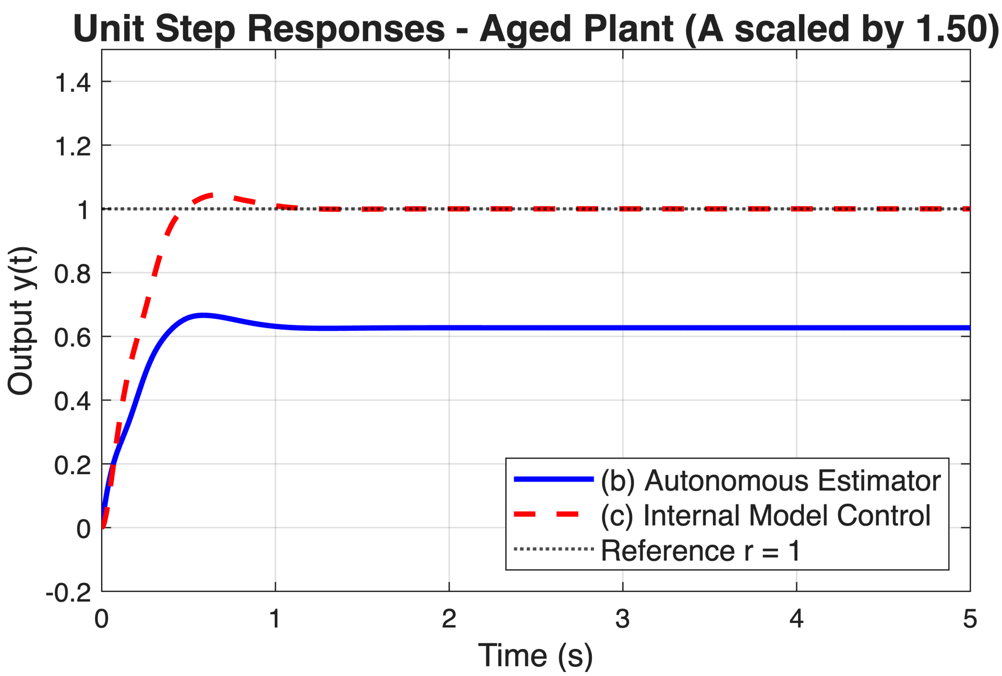
```
  -9.4829 +27.5555i
  -9.4829 -27.5555i
  -4.5527 + 4.4575i
  -4.5527 - 4.4575i
  -7.3720 + 4.4999i
  -7.3720 - 4.4999i
```

```
% --- (c) with aged plant: integrator + observer (designed for original A)
---
A_aug_c_aged = [ 0,         C,             zeros(1,n)  ;
                -B*K1,     A_aged,        -B*K0       ;
                -B*K1,     L*C,           A - L*C - B*K0 ];
B_aug_c_aged = [-1; zeros(n,1); zeros(n,1)];
C_aug_c_aged = [0, C, zeros(1,n)];
sys_c_aged = ss(A_aug_c_aged, B_aug_c_aged, C_aug_c_aged, 0);

fprintf('Part (c) aged poles:\n'); disp(eig(A_aug_c_aged));
```

```
Part (c) aged poles:
 -49.1491 + 0.0000i
 -34.9593 +25.2829i
 -34.9593 -25.2829i
  -5.5688 +23.6450i
  -5.5688 -23.6450i
  -5.0271 + 4.1378i
  -5.0271 - 4.1378i
  -6.8703 + 4.4175i
  -6.8703 - 4.4175i
```

```
% Stability check
if any(real(eig(Acl_b_aged)) >= 0) || any(real(eig(A_aug_c_aged)) >= 0)
    warning('One of the aged closed-loop systems is unstable. Try a smaller
aging factor.');
end

% Simulate
[y_b_aged, t_b_aged] = step(sys_b_aged, t);
[y_c_aged, t_c_aged] = step(sys_c_aged, t);

figure('Name','Part (e): Step Responses - Aged Plant (A x
1.5)','NumberTitle','off');
plot(t_b_aged, y_b_aged, 'b-',  'LineWidth', 2); hold on;
plot(t_c_aged, y_c_aged, 'r--', 'LineWidth', 2);
yline(1, 'k:', 'LineWidth', 1.2);
grid on;
xlabel('Time (s)', 'FontSize', 12);
ylabel('Output y(t)', 'FontSize', 12);
title(sprintf('Unit Step Responses - Aged Plant (A scaled by %.2f)',
aging_factor), 'FontSize', 14);
legend('(b) Autonomous Estimator', ...
       '(c) Internal Model Control', ...
       'Reference r = 1', ...
```

```
        'Location', 'southeast', 'FontSize', 11);
xlim([0 5]); ylim([-0.2 1.5]);
```




```
fprintf('Aged steady-state y_b(5s) = %.4f  (error = %.4f)\n',
y_b_aged(end), 1 - y_b_aged(end));
```

```
Aged steady-state y_b(5s) = 0.6269  (error = 0.3731)
```

```
fprintf('Aged steady-state y_c(5s) = %.4f  (error = %.4f)\n',
y_c_aged(end), 1 - y_c_aged(end));
```

```
Aged steady-state y_c(5s) = 1.0000  (error = 0.0000)
```